INCH-POUND

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5 February 1990
SUPERSEDING
WS 20623
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(See 6.6)

MILITARY SPECIFICATION

BROADGOODS, CARBON CLOTH, PHENOLIC-PREIMPREGNATED

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 **Scope.** This specification defines one type of carbon cloth broadgoods preimpregnated with phenolic resin, referred to herein as the cloth.

2. APPLICABLE DOCUMENTS

2.1 **Government documents.**

2.1.1 **Specifications, standards and handbooks.** The following specifications, standards and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (*DODISS*) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

MILITARY

MIL-R-9299	Resin, Phenolic, Laminating
MIL-C-82817	Carbon Reinforcement

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Ordnance Station, Standardization Branch (3730), Indian Head, MD 20640-5000, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 1340

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STANDARDS

MILITARY

MIL-STD-129 Marking for Shipment and Storage

(Unless otherwise indicated, copies of federal and military specifications, standards and handbooks are available from: Military Specifications and Standards, Bldg. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.2 **Non-Government publications.** The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM C 177 Steady-State Heat Flux Measurements and Thermal Transmission Properties by Means of the Guarded Hot Plate Apparatus (DoD adopted)

ASTM C 613 Resin Content of Carbon and Graphite Prepregs by Solvent Extraction (DoD adopted)

ASTM D 695 Compressive Properties of Rigid Plastics (DoD adopted)

ASTM D 740 Methyl Ethyl Ketone (DoD adopted)

ASTM D 790 Flexural Properties of Unreinforced and Reinforced Plastics and Electrical Insulating Materials (DoD adopted)

ASTM E 228 Linear Thermal Expansion of Solid Materials with a Vitreous Silica Delatometer (DoD adopted)

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103).

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 **Order of precedence.** In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.3) in accordance with 4.3.

3.2 Material. The cloth shall be woven carbon cloth broadgoods impregnated with a carbon-filled phenolic resin. The cloth shall be furnished in roll form, on a 3-inch diameter core, with all selvage removed.

3.2.1 Reinforcement. The reinforcement shall be a carbon fabric conforming to MIL-C-82817.

3.2.2 Resin filler. The resin filler shall be carbon black with a minimum carbon assay of 90 percent.

3.2.3 Resin. The resin system shall conform to MIL-R-9299, Grade A.

3.2.4 Roll width. The roll width shall be 40 inches minimum and 45 inches maximum.

3.3 Physical and mechanical properties.

3.3.1 Uncured material. The physical properties of the uncured cloth shall be in accordance with table I.

TABLE I. Uncured material physical properties.

Property	Percent by Weight	
	Minimum	Maximum
Cloth content	47.0	63.0
Resin solids, dry weight	32.0	37.0
Filler content	5.0	16.0
Volatile content	4.0	6.5
Resin flow	10.0	20.0

3.3.2 Cured material. The physical and mechanical properties of the cured cloth shall be in accordance with table II.

3.4 Age of material. The cloth shall be shipped within three months after the manufacturing date and shall be tested and certified within 60 days before shipment. The contractor shall certify that the cloth has been stored at a maximum temperature of 50°F between date of manufacture of the cloth and date of shipment. Temperature excursions above 50°F are permissible provided the cumulative time above 50°F does not exceed 30 days.

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TABLE II. Cured material physical and mechanical properties.

Property	Limits	
	Minimum	Maximum
Density, gram/cubic centimeter (g/cc)	1.44	1.55
Residual volatiles, percent (%)	0.0	2.5
Resin content, %	30.0	38.5
Compressive strength, edgewise, pounds per square inch (psi)		
Warp direction	30,000	65,000
Fill direction	20,000	55,000
Interlaminar double shear strength, psi	2,400	---
Thermal conductivity, Btu/(hr)(ft)(degrees Fahrenheit) (°F), 250°F, fill direction		
With ply	---	1.1
Across ply	---	0.9
Coefficient of thermal expansion at 75 to 400°F, in/in-°F x 10^{-6}		
Across ply	5.0	20.0
With ply	2.0	8.0
Flexural strength, psi		
Warp direction	25,000	55,000
Fill direction	20,000	50,000

3.5 **Long term storage.** The material shall be stored in watertight plastic bags at a maximum temperature of 50°F. Temperature excursions above 50°F are permissible provided the cumulative time above 50°F does not exceed 30 days.

3.6 **Shelf life.** The shelf life of the material shall be 9 months from date of manufacture when stored in watertight plastic bags at 50°F maximum.

3.6.1 **Shelf life extension.** The shelf life of cloth not being cut into tape may be extended for one additional six month period provided the average value of three specimens taken from the beginning of the cloth roll conforms to the following when tested as specified in 4.6.2.1 and 4.6.2.3.

	Minimum	Maximum
Resin flow, percent by weight	5.0	25.0
Volatile content, percent by weight	2.5	6.5

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3.7 **Workmanship.** The cloth shall have no holes, tears, non-uniform weave, resin build-up, contaminants, or other defects which would render more than 5% of any roll unsuitable for the intended use.

4. QUALITY ASSURANCE PROVISIONS

4.1 **Responsibility for inspection.** Unless otherwise specified in the contract or purchase order (see 6.2), the contractor is responsible for the performance of all inspection requirements (examinations and tests) specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 **Responsibility for compliance.** All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 **Classification of inspections.** The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).

b. Quality conformance inspection (see 4.4).

4.3 **First article inspection.** First article inspection shall consist of the tests specified in 4.4 and table III. Failure to meet all of the requirements specified herein shall be cause for rejection of the first article sample.

4.4 **Quality conformance inspection.** Quality conformance inspection shall consist of the examination and tests specified in table IV.

4.4.1 **Sampling for inspection.**

4.4.1.1 **Lot.** A lot shall consist of not more than 1000 pounds of cloth which has been impregnated in one continuous run using one batch of resin and filler, mixed in one vessel at one time, and offered for acceptance at one time. The weight of any single roll of cloth within the lot shall be 125 pounds maximum and 35 pounds minimum.

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TABLE III. First article inspection.

Tests	Requirement	Test Method
Cured Material		
Residual volatiles	3.3.2	4.6.3.1.2
Thermal conductivity	3.3.2	4.6.3.1.6
Coefficient of thermal expansion	3.3.2	4.6.3.1.7
Flexural strength	3.3.2	4.6.3.1.8
Compressive strength (fill)	3.3.2	4.6.3.1.4

TABLE IV. Quality conformance inspection.

Examination and Tests	Requirement	Test Method
Uncured material		
Visual examination	3.2, 3.7 and section 5	4.6.1
Volatile content test	3.3.1	4.6.2.1
Filler content test	3.3.1	4.6.2.2
Resin solids, dry weight test	3.3.1	4.6.2.2
Cloth content test	3.3.1	4.6.2.2
Resin flow test	3.3.1	4.6.2.3
Cured material		
Density test	3.3.2	4.6.3.1.1
Resin content test	3.3.2	4.6.3.1.3
Compressive strength (warp) test	3.3.2	4.6.3.1.4
Interlaminar shear test	3.3.2	4.6.3.1.5

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4.4.1.2 **Sampling for uncured properties.** Each roll in the lot shall be tested for uncured properties at the beginning and at the end of the roll. Three specimens from each location shall be tested.

4.4.1.3 **Sampling for cured properties.** Sampling for cured properties shall be in accordance with table V. The test specimens shall be removed from the beginning of the sample roll (see 6.4.2), and three specimens shall be tested.

TABLE V. Roll sampling for cured properties.

Number of Rolls in Lot	Number of Sample Rolls to be Selected for Testing
1 to 3	All
4 to 40	3

4.5 **Test results.** The average results of three tests at each location for each property shall conform to the specified property requirements in section 3. Individual specimen test results shall be recorded.

4.6 **Test methods.** The following test methods shall be used. Unless otherwise specified in the test, all weights, volumes, temperatures, and times shall be measured to the nearest specified unit or decimal.

NOTE: Reagent grade chemicals shall be used for chemical reactions in the conduct of all tests defined herein. Solvents and indicators may be commercial nonreagent grade materials.

4.6.1 **Visual examination.** Each container shall be visually examined for conformance to sections 3 and 5.

4.6.2 **Uncured material tests.**

4.6.2.1 **Volatile content test.**

a. Cut a 16 ± 2 square inch specimen with a minimum width of 1.5 inches, and weigh to nearest 0.001g (W1).

b. Place specimen in recirculating oven preheated to and stabilized at 325 ± 10°F for 10 ± 1 minutes. Specimens should be placed so they get adequate air circulation.

c. Remove specimen from oven, place in desiccator, and cool to room temperature.

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d. Remove specimen from desiccator, and weigh to nearest 0.001g (W2).

e. Calculate percent volatiles for each specimen as follows:

$$\text{Percent volatiles} = \frac{W1 - W2}{W1} \times 100$$

Where: W1 = original weight of specimen,g

W2 = final weight of specimen after devolatilization, g

f. Record average volatile content to nearest 0.1%.

4.6.2.2 **Resin solids content, filler content, and cloth content test.**

a. Place a 1.5 ± 0.1 g specimen weighed to the nearest 0.001 g (specimen weight) in 250 milliliter (ml) beaker.

b. Calculate devolatilized specimen weight (W) as follows:

$$W = \text{specimen weight} - \frac{(\text{percent volatiles})(\text{specimen weight})}{100}$$

Where: Percent volatiles is the value determined from the volatile content test (see 4.6.2.1).

c. Add 50 ml of N,N dimethyl formamide (DMF), and bring to boil. Gently stir boiling DMF and preimpregnated cloth for a minimum of 5 minutes. Remove beaker from heat source, and cool for a minimum of 5 minutes while stirring gently.

d. Decant and transfer liquid solution to second 250 ml beaker.

e. Rinse remaining fabric with DMF several times to ensure complete extraction of resin and filler from fabric, and add rinses to second 250 ml beaker.

f. Final rinse fabric with acetone to remove all traces of DMF.

g. Add all rinses to second 250 ml beaker.

h. Dry a filter and a filtering crucible at 350 ± 10°F for a minimum of 30 minutes. Weigh crucible and filter to nearest 0.001 g (W2).

i. Dry fabric in oven for 15 ± 1 minutes at 325 ± 10°F, cool in desiccator to room temperature, and reweigh to nearest 0.001 g (W1).

j. Using aspirator or vacuum pump, filter contents of second 250 ml beaker through filter and filtering crucible into vacuum flask.

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k. Rinse crucible, and filter several times with acetone to remove all resin.

l. Dry crucible and contents at 350 ± 10°F for 20 ± 1 minutes, cool in desiccator to room temperature, and reweigh to nearest 0.001 g (W3).

m. Calculate resin solids content, filler content, and cloth content as follows:

$$\text{Resin solids content (percent)} = \frac{W - (W1 + W4)}{W} \times 100$$

$$\text{Filler content (percent)} = \frac{W4}{W} \times 100$$

$$\text{Cloth content (percent)} = \frac{W1}{W} \times 100$$

Where: W = calculated devolatilized weight of preimpregnated specimen, g

W1 = weight of fabric after resin and filler removal, g

W2 = crucible and filter weight, g

W3 = filler + crucible and filter weight, g

W4 = W3 - W2 = filler content, g

n. Record resin solids content, filler content, and cloth content to nearest 0.1%.

4.6.2.2.1 **Alternate method for resin solids content, filler content, and cloth content test.** For dry resin solids content, carbon filler content, and cloth content, the requirements of 3.3.1 shall be verified in accordance with the Soxhlet extraction method specified in ASTM C 613.

4.6.2.3 **Resin flow test.**

a. Using a 4-inch by 4-inch ± 1/8-inch template, cut four squares across width of fabric. All squares shall be bias cut to eliminate fiber loss in testing. Stack squares uniformly on each other to make specimen.

NOTE: When retesting tape which is less than 4.0 inches for shelf life extension (see 3.6.1), it is permissible to use multiple tape widths which combine to provide a 4.0 inch specimen. Alternate tape layers shall be rotated 90°.

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b. Weigh specimen of four plies to nearest 0.01 g, and record as W1.

c. Place specimen between release film. Preheat press to 325 ± 10°F, position specimen in middle of press plate, and apply press load of 150 ± 10 pounds per square inch (psi). Press-load specimen for minimum of 10 minutes at 325 ± 10°F.

d. Remove specimen from press, and cool to ambient temperature.

e. Using knife, scrape off resin flash to original size of specimen. Do not remove any reinforcement from original dimensions.

f. Reweigh specimen to nearest 0.01 g, and record as W2.

g. Calculate percent resin flow as follows:

$$\text{Percent flow} = \frac{W1 - W2}{W1} \times 100$$

Where: W1 = original weight of specimen, g

W2 = final weight of specimen, g

h. Record resin flow to nearest 0.1 percent.

4.6.3 **Cured material tests.**

4.6.3.1 **Preparation of test panels.** Prepare test panels of sufficient size to provide the specimens for all required tests as follows:

a. Cut sample cloth as required to fabricate panel as specified in individual test procedures, and cure plies into a panel between separator film on smooth caul plates as follows:

(1) Orient plies such that the warp fibers of each ply are aligned in the same direction.

(2) Place stack of plies into press preset at 325 ± 10°F.

(3) Increase pressure slowly to 1000 ± 10 psi allowing resin to stage (see 6.4.4) to minimum flash at ply edges.

(4) Hold pressure and temperature for 120 ± 15 minutes, or 1 hour per inch of thickness, whichever is greater.

(5) Decrease temperature and pressure to ambient.

NOTE: The outer 1/2 inch of the test panel shall not be used for preparation of the specimens.

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4.6.3.1.1 Density test.

a. Cut sufficient plies to produce test panel 0.125 ± 0.025 by 4 inches (minimum) by 5 inches (minimum), mark warp direction, and prepare panel as specified in 4.6.3.1.

b. Cut specimen 1 inch by 1 inch by panel thickness at random from test panel. Wipe specimen clean using methyl ethyl ketone (MEK) conforming to ASTM D 740, and allow to dry in air a minimum of 20 minutes before testing.

c. Weigh and record weight of specimen in air to nearest 0.1 mg (W1).

d. Weigh specimen in water at 73 ± 3°F and record weight to nearest 0.1 mg, using wire and sinker if required (W2).

e. Calculate density as follows:

$$\text{Density (g/cc)} = \frac{0.9975\ W1}{(W1 + W3 - W2)}$$

Where: W1 = weight of specimen in air, g

W2 = weight of specimen, wire, and sinker suspended in water, g

W3 = weight of wire and sinker suspended in water to the same depth as in W2 determination to nearest 0.1 mg, g

0.9975 = conversion factor (specific gravity to density).

f. Record density to nearest 0.01 g/cc.

4.6.3.1.2 Residual volatile content test.

a. Cut sufficient plies to produce test panel 0.250 ± 0.050 inch thick by 4.000 ± 0.125 by 4.000 ± 0.125 inches. Prepare panel as specified in 4.6.3.1 with the following exception:
Apply contact pressure for 30 ± 5 seconds and dump pressure for 2 cycles before slowly applying maximum pressure. (The cloth shall not be staged.)

b. Cut specimen 1.000 ± 0.050 inch by 1.000 ± 0.050 inch by panel thickness from center 2-inch by 2-inch section of test panel. Top and bottom molded surfaces shall not be machined. Wipe specimen clean using MEK, and allow to air dry.

c. Place specimen in desiccator, and desiccate for 18 hours minimum.

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d. Weigh specimen to nearest 0.001 g, and record as W1.

e. Place specimen in air circulating oven preheated to and stabilized at 325 ± 10°F for 30 ± 6 minutes. Condition specimen at 325 ± 10°F for 4 to 4.5 hours.

f. Remove specimen from oven, and cool in desiccator for 30 ± 6 minutes or until specimen reaches room temperature.

g. Reweigh specimen to nearest 0.001 g. Record as W2.

h. Calculate percent residual volatiles as follows:

$$\text{Percent volatiles} = \frac{(W1 - W2)}{W1} \times 100$$

Where: W1 = original weight of specimen, g

W2 = devolatilized weight of specimen, g

h. Record residual volatiles to nearest 0.01%.

4.6.3.1.3 **Resin content test.** Resin content shall be determined in accordance with the following.

a. Prepare test specimens as follows:

(1) Cut 32 rectangles (1/2 inch by 5 inches nominal) from a prepreg sample.

(2) Stage the 32 rectangles at 255 ± 5°F to reduce flow.

NOTE: Staging time is dependent on the resin system and flow and must be determined from experience.

(3) Using the 32 rectangles, mold a bar 1/2 inch by 1/2 inch by 5 inches.

(4) Cure the specimen at 325 ± 10°F and 1000 ± 10 psi for 10 ± 1 minutes. If any resin flash, repeat and use longer staging time.

(5) Cut the bar into three equal lengths to form three test specimens.

b. Wipe specimen clean using MEK, and allow to dry in air 20 minutes minimum before testing.

c. Record specimen weight to nearest 0.001 g (W).

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d. Place test specimen in Vycor test tube, and tightly attach tube to test apparatus (see figure 1).

e. Start vacuum pump, and evacuate system to within 3 inches of mercury of absolute vacuum (minimum). Check for air leaks by clamping off hose to vacuum pump and noting manometer. The leak rate must be less than 0.2 inch of mercury per minute. Recheck joints until this level is achieved.

f. With vacuum on and pressure stabilized, place Fisher burner under the specimen. Adjust flame area to completely cover sample.

NOTE: A tube furnace maintained at 1500 ± 25 F may be used in place of the Fisher burners.

g. Continue pyrolysis for 30 ± 6 minutes or until manometer or gauge reading is same reading as it was before pyrolysis, whichever is longer.

NOTE: If residue collects in Vycor test tube, place Fisher burners under those areas until burnout is complete.

h. When pyrolysis is complete, turn off burners and cool under vacuum until specimen is at room temperature (touch Vycor test tube just under test specimen).

i. When specimen is cool, turn off vacuum and carefully let air into system. When pressure equilibrium is reached, remove specimen from test tube and weigh to nearest 0.001 g (W1).

$$\text{Resin content, percent} = \frac{(W - W1)}{W} \times 100 \times (K)$$

Where: W = weight of sample before pyrolysis, g

W1 = weight of sample after pyrolysis, g

K = constant ratio of percent resin (Soxhlet extraction of DMF wash) to percent weight loss by pyrolysis (K for U.S. Polymeric = 2.008, K for Fiberite = 2.060)

k. Record resin content to nearest 0.1%.

4.6.3.1.4 **Compressive strength test.** The compressive strength test shall be determined, using the test panel prepared in 4.6.3.1.1, in accordance with the following:

a. Machine specimen to dimensions shown on figure 2.

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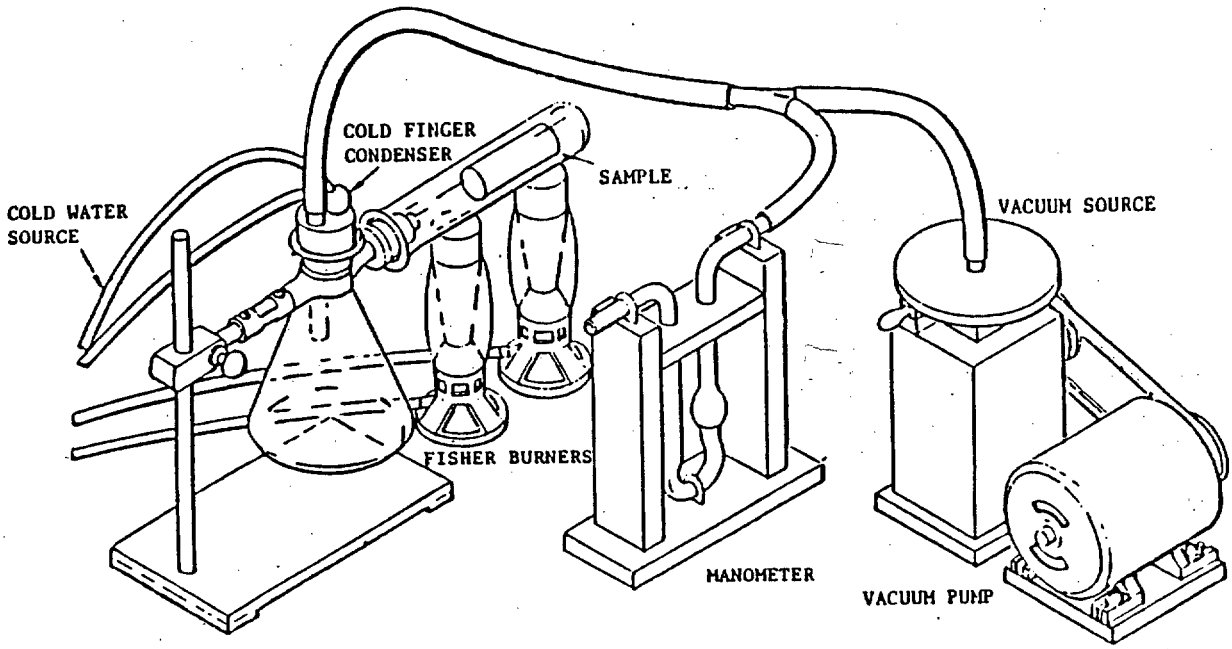


FIGURE 1. Resin content apparatus setup.

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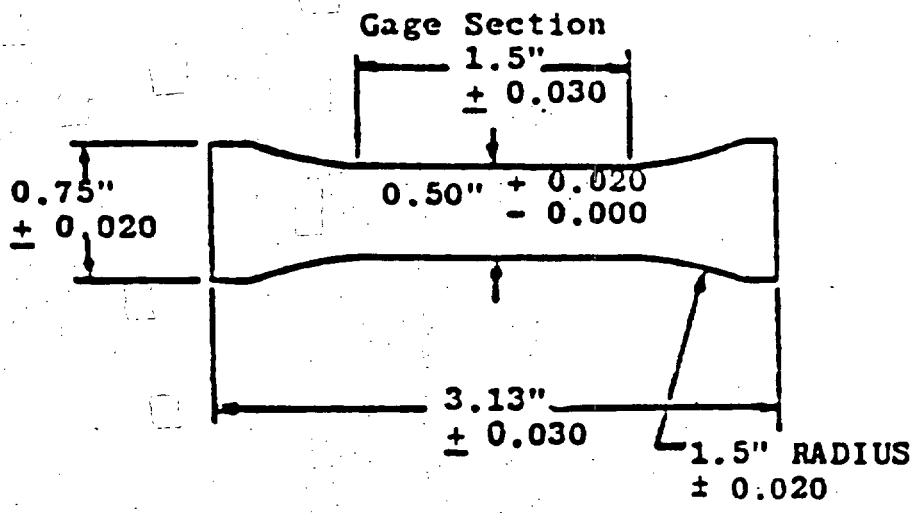


FIGURE 2. Compressive strength test specimen.

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b. Measure width and thickness of specimen to nearest 0.001 inch in gauge section area. Calculate and record cross-sectional area.

c. Place specimen in support jig (see figure 4 of ASTM D 695) centered so ends project an even amount beyond both ends of jig. Tighten finger tight only.

d. Place specimen and support jig in testing machine, and align so specimen ends are parallel with surface of compression tool. Adjust crosshead of testing machine until it just contacts top of specimen.

e. The testing machine crosshead speed shall be 0.05 ± 0.01 inch per minute. Start machine.

f. Record maximum load carried by specimen during test.

g. Calculate compressive strength by dividing maximum load by original cross-sectional area.

h. Record compressive strength to nearest 10 psi.

4.6.3.1.5 **Interlaminar shear test.**

a. Cut sufficient plies to produce cured layup panel 1.50 ± 0.25 inch thick across the plies by minimum dimensions of 1.5 by 1.5 inch.

b. Spray mold with release agent, and layup plies into mold.

c. Cure as specified in 4.6.3.1.

d. After cure, remove mold from press, remove test panel from mold, and allow panel to cool to room temperature.

e. Cut specimen of sufficient size to produce 0.375 inch diameter by 1.125 inch minimum length finished specimen, with length being perpendicular to plies from test panel.

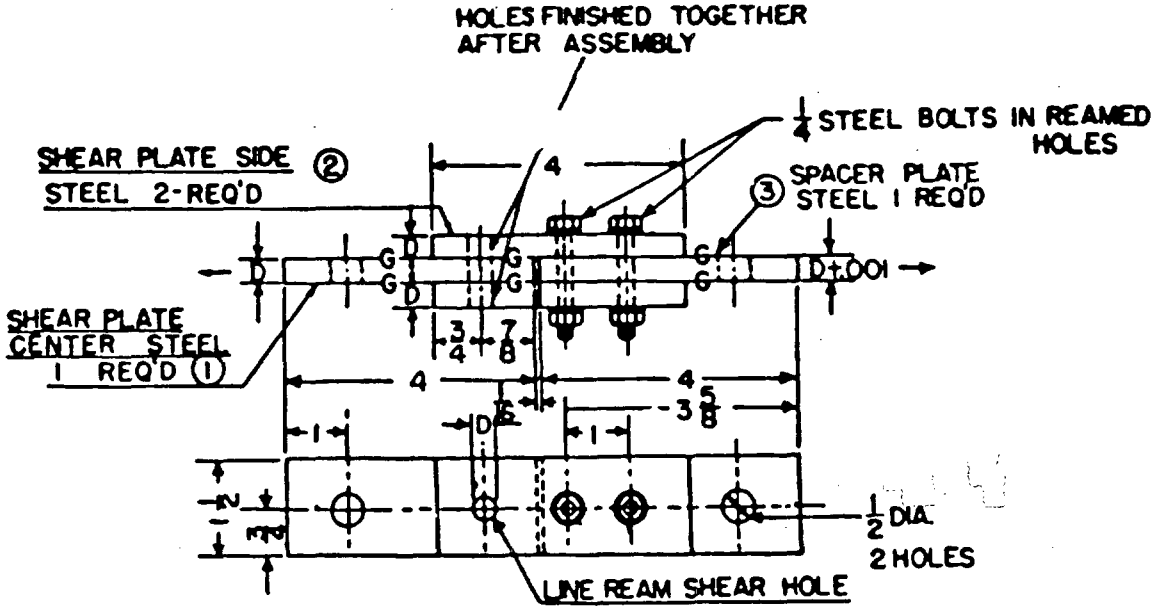
f. Center specimen to get length of cylindrical specimen perpendicular to plies, and machine specimens to diameter of 0.375 +0.000/-0.002 inch.

g. Measure diameter of test specimen to 0.0001 inch and calculate cross-sectional area (A).

h. Place specimen in a tensile loading shear jig conforming to figure 3.

i. Place shear fixture in testing machine, align, and pin ends of fixture.

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NOTES

1. D is nominal diameter of test cylinder.
2. Finish: machine after heat treatment.
3. Grind surfaces denoted G.
4. Spacer-plate (3) to be 0.001 inch thicker than (1).
5. Edges of shear hole must be sharp.
6. All dimensions in inches; tolerances—fractions ± 1/64; specimen diameter shall be D + 0.000, — 0.001; diameter shear hole shall be D + 0.000, — 0.001.
7. Alinement of shear holes must be maintained.
8. Material: steel, heat-treated to Rockwell C 35–40.

FIGURE 3. Shear jig, tensile loading.

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j. Set machine at indicated crosshead speed of 0.015 to 0.025 inch per minute, and test.

k. Record maximum load carried by specimen during test as L.

l. Calculations:

$$\text{Interlaminar shear (psi)} = \frac{L}{2A}$$

Where: L = maximum load carried by the specimen during testing, in pounds

A = original cross-sectional area of specimen, in square inches

m. Record interlaminar shear to nearest 10 psi.

4.6.3.1.6 **Thermal conductivity test.** Thermal conductivity shall be determined in accordance with ASTM C 177 and as follows:

a. Prepare test panel of required size, and cure as specified in 4.6.3.1.

b. Record data for temperature of 250 ± 10°F.

c. Maximum test panel size shall be 9.00 inches. Wrinkles are not permitted in the "with ply" specimen.

d. Record thermal conductivity to nearest 0.01 Btu/(hr) (ft) (°F).

4.6.3.1.7 **Coefficient of thermal expansion.** The coefficient of thermal expansion (see 6.4.1) shall be determined in accordance with ASTM E 228 and as follows:

a. Prepare test panel of required size and cure as specified in 4.6.3.1.

b. Data shall be recorded for temperature range from 75 to 400°F.

c. Maximum test panel specimen length shall be 4.0 inches.

d. Record coefficient of thermal expansion to nearest 0.01 in/in-°F x 10^{-6}.

4.6.3.1.8 **Flexural strength test.** Flexural strength shall be determined in accordance with ASTM D 790, Method I, and as follows:

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a. Prepare test panel of required size, and cure as specified in 4.6.3.1.

b. Maximum test specimen thickness shall be 0.25 inch.

c. Record flexural strength to nearest 10 psi.

4.7 **Inspection of packaging.** The cloth containers and container markings shall be visually examined for compliance with section 5.

5. PACKAGING

5.1 **Packaging.** Unless otherwise specified in the contract or order (see 6.2), packaging shall be level C as specified herein.

5.1.1 **Level C.** The cloth shall be packaged to afford adequate protection against loss, contamination, deterioration and damage during shipment from the supply source to the first receiving activity, and during storage under the shelf life period and conditions specified in 3.5 and 3.6. Containers in the same shipment shall be of the same size.

5.1.2 **Shipping container.** The cloth shall be shipped in a container that can be opened on the long side and will support both ends of the roll in the container.

5.2 **Marking.**

5.2.1 **Standard marking.** In addition to any special marking required by the contract or order, interior and exterior containers shall be marked in accordance with MIL-STD-129.

5.2.2 **Special marking.** In addition to the marking requirements of 5.2.1, each container marking shall include the following:

a. Title, number and date of this specification.

b. Manufacturer's name and address.

c. Date of manufacture.

d. Material trade name and identification.

e. Number of rolls.

f. Lot number and date of quality conformance inspection (see 4.4).

g. Shelf life (see 3.6) and inspection/test date. In addition, the marking shall include the following:

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"SHELF LIFE EXTENSION

THE SHELF LIFE OF AN INDIVIDUAL LOT MAY BE EXTENDED FOR AN ADDITIONAL 6 MONTH PERIOD PROVIDED THE CLOTH, UPON RETEST, CONFORMS TO THE RESIN FLOW AND VOLATILES CONTENT REQUIREMENT SPECIFIED IN MIL-B-82792."

h. Mark each roll on inside of spool with contractor name, roll number, lot number, and material name with ink stamp, stencil, or adhesive label.

i. Mark opposite sides of each shipping container with the following:

"TEMPERATURE SENSITIVE MATERIAL. DO NOT EXPOSE TO TEMPERATURE HIGHER THAN 100°F."

5.3 **Shipment temperature.** The temperature of the cloth during shipment shall not exceed 100°F as measured by a temperature-indicating tape placed on the inside of the spool of at least one cloth roll in each container in each shipment.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 **Intended use.** The cloth specified herein is intended for use in the manufacture of the Standard Missile Mk 104 Dual Thrust Rocket Motor nozzles.

6.2 **Acquisition requirements.** Acquisition documents must specify the following:

a. Title, number and date of this specification.

b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).

c. Whether first article inspection is required (see 3.1).

d. Packaging required, if other than as specified in 5.1.

e. Special marking, if other than as specified in 5.2.2.

6.3 **First article.** When first article inspection is required, the contracting officer should provide specific guidance to offerors on the material to be inspected. The cloth should be a first article sample and should consist of a quantity of the cloth sufficient to perform all of the inspections specified in 4.3. The first article sample should be packaged and marked as specified in section 5. The contracting officer should include specific

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instructions in acquisition documents regarding arrangements for examinations, approval of first article test results and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired to tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.4 Definitions.

6.4.1 Coefficient of thermal expansion. The secant slope between 297 and 478 degrees Kelvin of the linear expansion.

6.4.2 Sample roll. A cloth roll of impregnated cloth selected from the lot in accordance with the sampling plan (see 4.4.1.3).

6.4.3 Specimen. A portion of the sample cloth roll selected for inspection and test.

6.4.4 Stage. A method of stabilizing and retaining the phenolic polymer in a solid state on the carbon cloth by temperature cycling.

6.5 Subject term (key word) listing.

Nozzles, rocket
Rocket Motor, Dual Thrust, Mk 104
Standard Missile

6.6 Supersedure information. MIL-B-82792 incorporates the following engineering change proposals (ECPs), notice of revision (NOR) and specification change notice (SCN):

ECP	NOR	SCN
SSS1030 8/24/84		SCN 1 8/24/84
MTA062 3/30/88	MTA062.1 3/30/88	

Preparing Activity
NAVY-OS
(Project 1340-N799)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-B-82792	2. DOCUMENT DATE (YYMMDD) 5 February 1990

3. DOCUMENT TITLE
BROADGOODS, CARBON CLOTH, PHENOLIC-PREIMPREGNATED

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME *(Last, First, Middle Initial)*	b. ORGANIZATION	
c. ADDRESS *(Include Zip Code)*	d. TELEPHONE *(Include Area Code)* (1) Commercial (2) AUTOVON *(If applicable)*	7. DATE SUBMITTED *(YYMMDD)*

8. PREPARING ACTIVITY NAVAL ORDNANCE STATION (CODE 3730) INDIAN HEAD, MD 20640-5000

a. NAME	b. TELEPHONE *(Include Area Code)* (1) Commercial (2) AUTOVON
c. ADDRESS *(Include Zip Code)*	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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